WORKED SOLUTION

STEP III 2016, Question 7

Pure • Roots of Unity

STEP III 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 5.4, *Distances in a Regular Polygon*: https://beyondstep.keithmcnulty.org/ch05/polygon-distances.html

The key idea

Distances are moduli, and a product of moduli is the modulus of a product. If we put the polygon in the Argand diagram so that its vertices are rotated nth roots of unity, the product of distances becomes $|w^n - 1|$ for a suitable w, by the opening identity. The only real work is choosing the coordinates well: put O at the origin and make the perpendicular bisector of X_0X_1 the real axis, so that P is a real number.

Solution

The factorisation

For $k = 0, 1, \ldots, n-1$ we have $(\omega^k)^n = \mathrm{e}^{2\pi \mathrm{i} k} = 1$, so each ω^k is a root of $z^n - 1$. These n numbers are distinct, because their arguments $2\pi k/n$ are distinct numbers in $[0, 2\pi)$. Both sides of the identity are monic polynomials of degree n, so their difference

$$D(z) = (z-1)(z-\omega)\cdots(z-\omega^{n-1}) - (z^n - 1)$$

has degree at most $n-1$. But D vanishes at the n distinct points $1, \omega, \ldots, \omega^{n-1}$, and a non-zero polynomial of degree at most $n-1$ has at most $n-1$ roots. Hence $D \equiv 0$, and

$$\prod_{k=0}^{n-1}(z - \omega^k) = z^n - 1 \quad \text{for every complex } z.$$

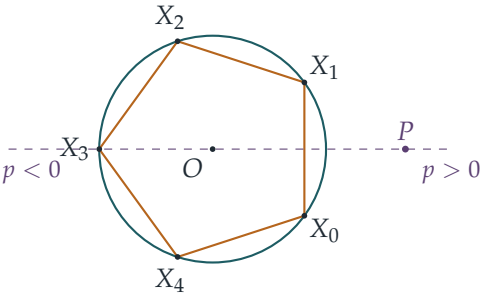
Part (i)

We take the vertices to be labelled in order round the circle, so X_0 and X_1 are adjacent and $\angle X_0 O X_1 = 2\pi/n$. Rotating or reflecting the figure changes no distances, so we

may place O at the origin with

$$X_k = \mathrm{e}^{\mathrm{i}(2k-1)\pi/n} = \mathrm{e}^{-\mathrm{i}\pi/n}\,\omega^k, \qquad k = 0, 1, \ldots, n-1.$$

Then $X_0 = \mathrm{e}^{-\mathrm{i}\pi/n}$ and $X_1 = \mathrm{e}^{\mathrm{i}\pi/n}$ are reflections of each other in the real axis, so the perpendicular bisector of X_0X_1 is the real axis. The points equidistant from X_0 and X_1 are exactly the points of this line, so P is a real number p, and $|OP| = |p|$. The midpoint of X_0X_1 is $\cos(\pi/n) > 0$, so $p > 0$ means P is on the same side of O as the side X_0X_1.



(The figure shows $n = 5$; for odd n the line through O also passes through the opposite vertex.)

Now, using $|ab| = |a||b|$ and $|\mathrm{e}^{-\mathrm{i}\pi/n}| = 1$,

$$\begin{aligned}|PX_0|\,|PX_1|\cdots|PX_{n-1}| &= \prod_{k=0}^{n-1}|p - \mathrm{e}^{-\mathrm{i}\pi/n}\omega^k| = \prod_{k=0}^{n-1}|\mathrm{e}^{-\mathrm{i}\pi/n}|\,|p\,\mathrm{e}^{\mathrm{i}\pi/n} - \omega^k| \\ &= \left|\prod_{k=0}^{n-1}(p\,\mathrm{e}^{\mathrm{i}\pi/n} - \omega^k)\right| = \left|(p\,\mathrm{e}^{\mathrm{i}\pi/n})^n - 1\right|,\end{aligned}$$

by the factorisation with $z = p\,\mathrm{e}^{\mathrm{i}\pi/n}$. Since $\mathrm{e}^{\mathrm{i}\pi} = -1$,

$$|PX_0|\,|PX_1|\cdots|PX_{n-1}| = |-p^n - 1| = |p^n + 1|.$$

Ask yourself Why does the sign of p matter for odd n but not for even n?

If n is even, then $p^n = |p|^n \geq 0$ whatever the sign of p, so $|p^n + 1| = |p|^n + 1$, and

$$|PX_0| \times |PX_1| \times \cdots \times |PX_{n-1}| = |OP|^n + 1,$$

as required. If n is odd, then p^n has the same sign as p, and there are two cases.

- If P lies on the ray from O through the midpoint of X_0X_1 ($p \geq 0$), then $p^n = |OP|^n$ and the product is $|OP|^n + 1$.
- If P lies on the opposite ray ($p < 0$), which for odd n is the ray from O through the vertex $X_{(n+1)/2}$ (at $\mathrm{e}^{\mathrm{i}\pi} = -1$), then $p^n = -|OP|^n$ and the product is

$$\big|\,|OP|^n - 1\big|.$$

The modulus signs matter in the second case: the value is $1 - |OP|^n$ when P is inside the circle, and it is 0 when $P = X_{(n+1)/2}$, as it must be, since one of the distances is then zero.

Part (ii)

This time rotate so that $X_k = \omega^k$, so $X_0 = 1$. For $z \neq 1$ the factorisation gives

$$(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}) = \frac{z^n-1}{z-1} = 1 + z + z^2 + \cdots + z^{n-1}.$$

Both ends are polynomials that agree at infinitely many points (every $z \neq 1$), so they are equal for all z, including $z = 1$. Putting $z = 1$:

$$(1-\omega)(1-\omega^2)\cdots(1-\omega^{n-1}) = \underbrace{1+1+\cdots+1}_{n} = n.$$

Taking moduli, and using $|1-\omega^k| = |X_0X_k|$,

$$|X_0X_1| \times |X_0X_2| \times \cdots \times |X_0X_{n-1}| = n,$$

which is the required result.

Checks

- Square ($n = 4$) with $P = O$: every distance is 1, so the product is $1 = 0^4 + 1$. With P far away, the product behaves like $|OP|^n$, as the formula says.
- Equilateral triangle ($n = 3$) with P on the far side at distance 2 from O: the distances are 1 (to the opposite vertex at -1) and $\sqrt{(-2-\frac{1}{2})^2 + \frac{3}{4}} = \sqrt{7}$ twice, giving $7 = 2^3 - 1$.
- Part (ii) for the square: $|X_0X_1|\,|X_0X_2|\,|X_0X_3| = \sqrt{2} \times 2 \times \sqrt{2} = 4$. All three results were also confirmed by computer for random n, rotations and positions of P.

Common slips

- Placing X_0 at 1 in part (i): then the perpendicular bisector of X_0X_1 is the line at angle π/n, not the real axis, and the algebra is messier. Choose coordinates to suit P.
- Forgetting the modulus in the odd case with P on the far side: $|OP|^n - 1$ is negative when P is inside the circle.
- Substituting $z = 1$ straight into $\dfrac{z^n-1}{z-1}$; the polynomial identity is what justifies the value n.

Beyond STEP

Part (ii) is the statement that the derivative of $z^n - 1$ at $z = 1$ is n: in general, if $f(z) = \prod(z - a_k)$ then $f'(a_0) = \prod_{k \neq 0}(a_0 - a_k)$. Products over roots of unity such as these appear in the theory of cyclotomic polynomials and in computing discriminants, and part (i) is essentially a classical result of Roger Cotes on distances from a point on a line of symmetry of a regular polygon. Needham's *Visual Complex Analysis* is a beautiful place to see this geometry of complex multiplication developed.

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